ASCOMP 2026 POSTER

Min-Marginal Power Indices for Multi-Objective Feature Selection

SCAN ME

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01 PRINCIPLE & TOOLKIT

A coalition witness reveals harmful additions: acceptable utility becomes unacceptable.

N: analyzed features; $U_{val}(N) > 0$; $0 < \alpha \le 1$

$$v_\alpha(S) = 1[U_{val}(S) \ge \alpha\, U_{val}(N)]$$

$$\Delta_i(S) = v_\alpha(S \cup \{i\}) - v_\alpha(S)$$

$$\mathbf{MinPI_i = MinSS_i = MinBz_i = \min \Delta_i(S)}$$

Minimum over $S \subseteq N \setminus \{i\}$; exact labels:

-1 harmful: some coalition flips 1 → 0.
0 neutral: the minimum marginal is 0.
+1 decisive: every coalition flips 0 → 1.

MinSS / MinBz: minimum Shapley-Shubik / Banzhaf.
Same support; exact enumeration uses $\le 2^n$ utilities.

CORE TOOLS

MI (mutual information) ranks feature-target dependence.
A **maximum-weight absolute-correlation tree** represents pairwise association and guides Algorithm B.

XGBoost: 100 rounds, depth 4, learning rate 0.1; CUDA trees. Fit coalition models on training data.

Training-only preprocessing: variance filtering → MI selection (≤100 features) → standardization.

Utility: ROC AUC / one-vs-rest AUC / R^2.
Empty coalition: training class priors / response mean.

MATCHED-COUNT CONTROLS

All controls delete the accepted MinPI count k.
SHAP-bottom: lowest mean |TreeSHAP| on train.
MinShap / Banzhaf-bottom: lowest minimum / mean continuous utility marginal from shared coalitions.
Random: 30 matched-count trials per seed.

α = 0.95; quota sweep: 0.85, 0.90, 0.95, 0.99.

02 ALGORITHM FLOW

Train 64% → validation 16% → test 20%
Fit on train; compute the threshold game on validation.

Algorithm A: size-uniform sampling
Draw s uniformly from {0, ..., n-1}; then draw an s-feature subset uniformly from $N \setminus \{i\}$.

Algorithm B: MI-correlation guidance
B_i: top ≤10 tree neighbors by absolute correlation;
pad from the top-MI half of $N \setminus \{i\}$ toward min(20, floor(n/3)).
Each draw: reset to B_i; drop each independently (p = 0.3).
Draw a uniformly from 0, ..., min(5, n-1);
add *a* absent features from $N \setminus \{i\}$ in random order.

Both: $m \leftarrow \min(m, \Delta_i(S))$, initialized at +1.
Stop at -1 with a witness; else return m after K draws.
K = {2, 5, 10, 25} per feature; ISOLET up to 10.
Removal uses the largest K.

HYBRID DISPATCH
n ≤ 20: exact | 20 < n ≤ 75: B* | n > 75: min(A, B)
*A fallback without correlation guidance; minimum is featurewise.

Validate joint removal
H = {i: estimated $MinPI_i = -1$}

$$\mathbf{U_{val}(N \setminus H) \ge U_{val}(N) - \varepsilon}$$

ε = 0.005 (classification); 0.01 (regression).
PASS: remove all H. **FAIL:** keep all N.
Output retained features; evaluate on held-out test data.

Optional MinPI-prefiltered NSGA-II
Non-dominated sorting genetic algorithm II; minimize:

$$\mathbf{(|S|/n,\ 1 - clip(U_{val}(S), 0, 1))}$$

n is shared: analyzed dimension before prefiltering.
population 40; 800 evaluations per variant.
HV reference (1.1, 1.1); select fronts on validation.
Evaluate the selected fronts on held-out test data.

03 EXPERIMENTAL EVIDENCE

6 datasets × 5 seeds

Seeds: 42, 123, 456, 789, 1024 | default α = 0.95

12.53%	-0.000313
mean feature removal	mean held-out ΔU

Five-seed mean ± SD. ΔU = retained - full utility.
Removal of analyzed features; bars use a 0-80% scale.

DATASET	REMOVED (%)	HELD-OUT ΔU
Madelon	43.60 ± 31.96	+0.00469 ± 0.00497
Abalone	27.50 ± 25.62	-0.00565 ± 0.01077
Credit-Card-Fraud	2.76 ± 2.89	-0.00078 ± 0.00335
Spambase	0.72 ± 0.99	-0.00003 ± 0.00007
Gisette	0.60 ± 0.89	-0.00011 ± 0.00016
ISOLET	0.00 ± 0.00	0.00000 ± 0.00000

MATCHED-COUNT REMOVAL: MinPI vs random
Madelon: ΔU +0.0047 vs -0.0732
Abalone: ΔU -0.0056 vs -0.0370

NSGA-II: held-out hypervolume

ΔHV = MinPI-prefiltered - full-space search.
Mean +0.01584; Madelon +0.09515; Holm p = 0.4375.
Positive dataset-averaged change in all five seeds.

ESTIMATOR ACCURACY

Abalone (8 features): harmful-set F1 vs exact MinPI.
Five-seed mean; hybrid / exact F1 = 1 at every K.

K	2	5	10	25
A	0.100	0.150	0.230	0.443
B	0.000	0.080	0.133	0.313